Accelerated solutions of convection-dominated PDEs

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Abstract

This work introduces a hyperreduction procedure [1] to effectively reduce the computational cost of solving convection-dominated problems. The reduced-order model is defined as the solution of a residual minimization problem over the nonlinear manifold. An online-efficient method is obtained by using empirical quadrature to approximate the optimality system such that it can be solved with mesh-independent operations.

1. Governing equations

Consider a $\boldsymbol{\mu}$-parametrized system of m conservation laws in d-dimensions

$$\nabla \cdot f(u, \nabla u; \boldsymbol{\mu}) = s(u, \nabla u; \boldsymbol{\mu}) \quad \text{in} \quad \Omega$$

where $\boldsymbol{\mu} \in \mathcal{D}$ is the parameter domain, $u(x; \boldsymbol{\mu}) \in \mathbb{R}^m$ is the solution at a point $x \in \Omega$, $f : \mathbb{R}^m \times \mathbb{R}^{m\times d} \times \mathcal{D} \to \mathbb{R}^{m\times d}$ is the flux function, and $s : \mathbb{R}^m \times \mathbb{R}^{m\times d} \times \mathcal{D}^m \to \mathbb{R}^m$ is the source term. The PDE is discretized to yield a nonlinear system of algebraic equations

$$\boldsymbol{R}(\boldsymbol{U}; \boldsymbol{x}, \boldsymbol{\mu}) = \sum_{e=1}^{N_e} \boldsymbol{P}_e^T \boldsymbol{R}_e(\boldsymbol{P}_e \boldsymbol{U}, \hat{\boldsymbol{P}}_e \boldsymbol{U}; \tilde{\boldsymbol{P}}_e \boldsymbol{x}, \boldsymbol{\mu}),$$

where $\boldsymbol{U}(\boldsymbol{x}, \boldsymbol{\mu}) \in \mathbb{R}^N$ is the solution coefficients of the discretized PDE as the high-dimensional model (HDM) $\boldsymbol{x}(\boldsymbol{\mu}) \in \mathbb{R}^{\tilde{N}}$ is the coefficients of the domain mapping, $\boldsymbol{R} : \mathbb{R}^N \times \mathbb{R}^{\tilde{N}} \times \mathcal{D} \to \mathbb{R}^N$ denotes the nonlinear residual function, $\boldsymbol{R}_e : \mathbb{R}^N \times \mathbb{R}^{\tilde{N}} \times \mathcal{D} \to \mathbb{R}^{n_e}$ is the contribution of the mesh entity e to the global, and $\boldsymbol{P}_e \in \mathbb{R}^{n_e \times N}, \hat{\boldsymbol{P}}_e \in \mathbb{R}^{\hat{n}_e \times N}, \tilde{\boldsymbol{P}}_e \in \mathbb{R}^{\tilde{n}_e \times \tilde{N}}$ are the Boolean column-orthogonal matrices that localizes global vectors.

2. Model reduction with implicit feature tracking

Central to the implicit feature tracking method introduced in [2] is the following low-dimensional linear approximation

$$\boldsymbol{U}^\star\left(\hat{\boldsymbol{x}}^\star_{\boldsymbol{\Phi},\Psi}(\boldsymbol{\mu}), \boldsymbol{\mu}\right) \approx \hat{\boldsymbol{U}}^\star_{\boldsymbol{\Phi},\Psi}(\boldsymbol{\mu}) := \boldsymbol{\Phi}\hat{\boldsymbol{w}}^\star_{\boldsymbol{\Phi},\Psi}(\boldsymbol{\mu}), \quad \hat{\boldsymbol{x}}^\star_{\boldsymbol{\Phi},\Psi}(\boldsymbol{\mu}) := \boldsymbol{X} + \boldsymbol{\Psi}\hat{\boldsymbol{\tau}}^\star_{\Phi,\Psi}(\boldsymbol{\mu}),$$

where $\hat{\boldsymbol{x}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : \mathcal{D} \to \mathbb{R}^{N_{\boldsymbol{x}}}$ is a low-dimensional parametrization of the coefficients $\boldsymbol{x}$ with reduced basis $\boldsymbol{\Psi} \in \mathbb{R}^{N_{\boldsymbol{u}} \times k_{\boldsymbol{x}}}$ and coefficients $\hat{\boldsymbol{\tau}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : \mathcal{D} \to \mathbb{R}^{k_{\boldsymbol{x}}}$ that leads to $(k_{\boldsymbol{x}} \ll N_{\boldsymbol{x}})$ $\hat{\boldsymbol{U}}^\star_{\boldsymbol{\Phi},\Psi} : \mathcal{D} \to \mathbb{R}^{N_u}$ is the low-dimensional approximation of the state with reduced basis $\boldsymbol{\Phi} \in \mathbb{R}^{N_u \times k_u}$ and coefficients $\hat{\boldsymbol{w}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : \mathcal{D} \to \mathbb{R}^{k_u}$ $(k_{\boldsymbol{u}} \ll N_{\boldsymbol{u}})$. The reduced coefficients are determined simultaneously as the solution of the optimization problem

$$\left(\hat{\boldsymbol{w}}^\star_{\boldsymbol{\Phi},\Psi}(\boldsymbol{\mu}), \hat{\tau}^\star_{\boldsymbol{\Phi},\Psi}(\boldsymbol{\mu})\right) := \underset{(\boldsymbol{w},\boldsymbol{\tau}) \in \mathbb{R}^{k_u} \times \mathbb{R}^{k_{\boldsymbol{x}}}}{\arg\min} \; \hat{J}_{\boldsymbol{\Phi},\Psi}(\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu}),$$

where the objective function $\hat{J}_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : \mathbb{R}^{k_u} \times \mathbb{R}^{k_{\boldsymbol{x}}} \times \mathcal{D}$ is defined as the norm of the residual function $\boldsymbol{F} : \mathbb{R}^{N_u} \times \mathbb{R}^{N_{\boldsymbol{x}}} \times \mathcal{D} \to \mathbb{R}^{N_u + N_e}$ that incorporates the HDM residual and a mesh distortion term

$$\hat{J}_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : (\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu}) \mapsto \frac{1}{2}\|\boldsymbol{F}(\boldsymbol{\Phi}\boldsymbol{w}, \boldsymbol{X} + \boldsymbol{\Psi}\boldsymbol{\tau}; \boldsymbol{\mu})\|_2^2, \quad \boldsymbol{F} : (\boldsymbol{U}, \boldsymbol{x}; \boldsymbol{\mu}) \mapsto \begin{bmatrix} \boldsymbol{R}(\boldsymbol{U}; \boldsymbol{x}, \boldsymbol{\mu}) \\ \boldsymbol{N}(\boldsymbol{x}) \end{bmatrix},$$

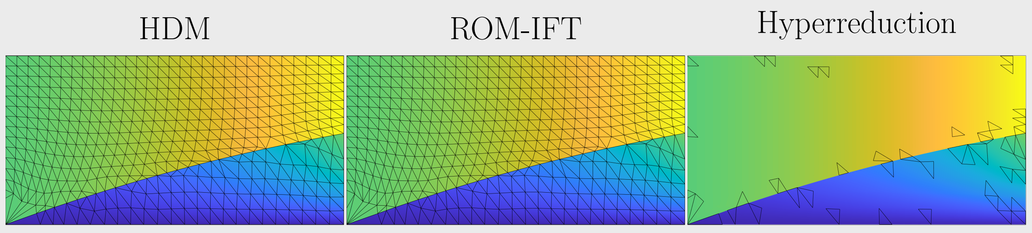
where the residual $N : \mathbb{R}^{N_{\boldsymbol{x}}} \to \mathbb{R}^{N_e}$ is a measure of mesh distortion.

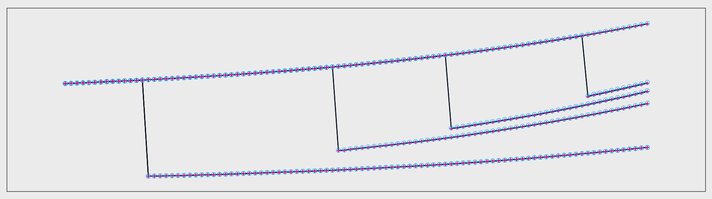
4. Inviscid Burgers equation

Consider the time-dependent inviscid Burgers' equations in the space-time domain $\Omega := (0, 100) \times (0, 50)$

$$\partial_t u(x, t; \boldsymbol{\mu}) + u(x, t; \boldsymbol{\mu})\partial_x u(x, t; \boldsymbol{\mu}) = g(x; \boldsymbol{\mu}) \quad (x, t) \in \Omega$$

where $u(x, t; \boldsymbol{\mu}) \in \mathbb{R}$ is implicitly defined as the solution, the parameter vector is $\boldsymbol{\mu} \in \mathcal{D} := [3, 9] \times [0.02, 0.075]$ and the source term $g : \Omega \times \mathcal{D} \to \mathbb{R}$





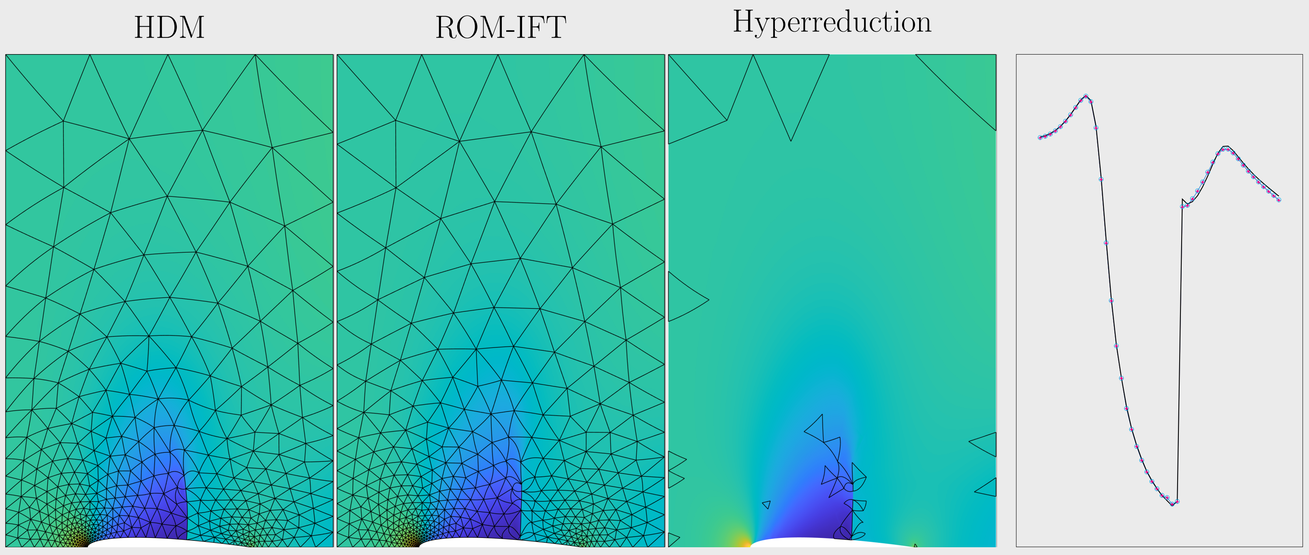
Solution of the inviscid Burgers equation problem at $\boldsymbol{\mu} = (4.5, 0.038)$ (1st row) and along a five temporal slices $\{(s, 5t) \mid s \in (0, 100), t = 1, \ldots, 6\}$ (2nd row). Legend: HDM (–), ROM-IFT(→), and ROM-IFT hyperreduction(→).

5. Transonic flow over NACA0012

Consider the inviscid, transonic flow of an ideal gas over a NACA0012 airfoil modeled by the compressible Euler equations. Let $\Omega \subset \mathbb{R}^2$ denote the flow domain, then the Euler equations can be written as an inviscid conservation law with solution $u : \Omega \times \mathcal{D} \to \mathbb{R}^4$ and components $u = (\rho, \rho v, \rho E)$, where $\rho : \Omega \times \mathcal{D} \to \mathbb{R}_{>0}, v : \Omega \times \mathcal{D} \to \mathbb{R}^d, E : \Omega \times \mathcal{D} \to \mathbb{R}_{>0}$ are the density, velocity, and energy of the fluid, respectively. The flux function and source term are defined as

$$f(u) := \begin{bmatrix} \rho v^T \\ \rho v \otimes v + P(u) I_d \\ (\rho E + P(u)) v^T \end{bmatrix}, \quad s(u) := 0.$$

The freestream Mach number $M_\infty \in [0.75, 0.82]$ is used to parametrize the problem.



Density Solution of the transonic flow problem at Mach=0.78 (1st three left) and a horizontal slice along the curve $\{(s, 0.14) | s \in (-0.5, 1.5)\}$ (right). Legend: HDM (–), ROM-IFT(→), and ROM-IFT Hyperreduction(→).

3. Empirical quadrature procedure for implicit feature tracking

Let $\check{\boldsymbol{\rho}} \in \mathbb{R}^{N_e}$ be a guess to the sparse weights which define $\boldsymbol{\rho}^\star_{\Phi,\Psi,\check{\rho}}$ as the solution of the following linear program

$$\boldsymbol{\rho}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}} := \underset{\boldsymbol{\rho} \in \mathcal{C}_{\boldsymbol{\Phi},\Psi,\check{\rho}}}{\arg\min} \sum_{e=1}^{N_e} \rho_e, \quad \mathcal{C}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}} := \mathcal{C}^{\text{nn}} \cap \mathcal{C}^{\text{dv}} \cap \mathcal{C}^{\text{sr}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}} \cap \mathcal{C}^{\text{dr}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}}$$

where the constraints are defined as

$$\mathcal{C}^{\text{nn}} := \left\{\boldsymbol{\rho} \in \mathbb{R}^{N_e} \mid \rho_e \geqslant 0, e = 1, \ldots, N_e\right\}$$

$$\mathcal{C}^{\text{dv}} := \left\{\boldsymbol{\rho} \in \mathbb{R}^{N_e} \middle| \, |\Omega_0| - \sum_{e=1}^{N_e} \rho_e \, |\Omega_{0,e}| \, | \leqslant \delta |\Omega_0| \right\}$$

$$\mathcal{C}^{\text{sr}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}} := \left\{\boldsymbol{\rho} \in \mathbb{R}^{N_e} \middle| \left\| \hat{\boldsymbol{S}}_{\boldsymbol{\Phi},\boldsymbol{\Psi}}\left(\tilde{\boldsymbol{v}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}}(\boldsymbol{\mu}); \boldsymbol{\mu}\right) - \tilde{\boldsymbol{S}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\boldsymbol{\rho}}\left(\tilde{\boldsymbol{v}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}}(\boldsymbol{\mu}); \boldsymbol{\mu}\right) \right\|_\infty \leqslant \delta \left|\boldsymbol{\Xi}\right|^{-1}, \boldsymbol{\mu} \in \boldsymbol{\Xi}\right\}$$

$$\mathcal{C}^{\text{dr}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}} := \left\{\boldsymbol{\rho} \in \mathbb{R}^{N_e} \middle| \left\| \hat{\boldsymbol{T}}_{\boldsymbol{\Phi},\boldsymbol{\Psi}}\left(\tilde{\boldsymbol{v}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}}(\boldsymbol{\mu}); \boldsymbol{\mu}\right) - \tilde{\boldsymbol{T}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\boldsymbol{\rho}}\left(\tilde{\boldsymbol{v}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}}(\boldsymbol{\mu}); \boldsymbol{\mu}\right) \right\|_\infty \leqslant \delta \left|\boldsymbol{\Xi}\right|^{-1}, \boldsymbol{\mu} \in \boldsymbol{\Xi}\right\},$$

and $\tilde{\boldsymbol{v}}^\star_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}}(\boldsymbol{\mu}) := (\tilde{\boldsymbol{w}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\boldsymbol{\rho}}}(\boldsymbol{\mu}), \tilde{\boldsymbol{\tau}}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\check{\rho}}(\boldsymbol{\mu}))$ is introduced for notational brevity, $\delta \in \mathbb{R}_{>0}$ is a user-defined tolerance, and $\boldsymbol{\Xi} \subset \mathcal{D}$ is a collection of parameters used to train the weights where

$$\hat{\boldsymbol{S}}^{(e)}_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : (\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu}) \mapsto \left(\boldsymbol{\Phi}_e^T \frac{\partial \boldsymbol{R}_e}{\partial \boldsymbol{U}_e}\left(\hat{\boldsymbol{U}}_e, \hat{\boldsymbol{U}}'_e, \hat{\boldsymbol{x}}_e, \boldsymbol{\mu}\right)^T + (\boldsymbol{\Phi}'_e)^T \frac{\partial \boldsymbol{R}_e}{\partial \boldsymbol{U}'_e}\left(\hat{\boldsymbol{U}}_e, \hat{\boldsymbol{U}}'_e, \hat{\boldsymbol{x}}_e, \boldsymbol{\mu}\right)^T\right) \boldsymbol{R}_e\left(\hat{\boldsymbol{U}}_e, \hat{\boldsymbol{U}}'_e, \hat{\boldsymbol{x}}_e, \boldsymbol{\mu}\right)$$

$$\hat{\boldsymbol{T}}^{(e)}_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : (\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu}) \mapsto \boldsymbol{\Psi}_e^T \left(\frac{\partial \boldsymbol{R}_e}{\partial \boldsymbol{x}_e}\left(\hat{\boldsymbol{U}}_e, \hat{\boldsymbol{U}}'_e, \hat{\boldsymbol{x}}_e, \boldsymbol{\mu}\right)^T \boldsymbol{R}_e\left(\hat{\boldsymbol{U}}_e, \hat{\boldsymbol{U}}'_e, \hat{\boldsymbol{x}}_e, \boldsymbol{\mu}\right) + \frac{\partial \boldsymbol{N}_e}{\partial \boldsymbol{x}_e}(\hat{\boldsymbol{x}}_e)^T \boldsymbol{N}_e(\hat{\boldsymbol{x}}_e)\right)$$

$$\tilde{\boldsymbol{S}}^{(e)}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\rho} : (\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu}) \mapsto \sum_{e=1}^{N_e} \rho_e \hat{\boldsymbol{S}}^{(e)}_{\boldsymbol{\Phi},\boldsymbol{\Psi},\rho} : (\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu}), \quad \tilde{\boldsymbol{T}}^{(e)}_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : (\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu}) \mapsto \sum_{e=1}^{N_e} \rho_e \hat{\boldsymbol{T}}^{(e)}_{\boldsymbol{\Phi},\boldsymbol{\Psi}} : (\boldsymbol{w}, \boldsymbol{\tau}; \boldsymbol{\mu})$$

References

[1] Marzieh Alireza Mirhoseinia Matthew J. Zahr. "Accelerated solutions of convection-dominated partial differential equations using implicit feature tracking and empirical quadrature". In: *arxiv: 2305.15661* (2023). URL: `arxiv:%202305.15661` (visited on 05/25/2023).

[2] Marzieh Alireza Mirhoseini and Matthew J. Zahr. "Model reduction of convection-dominated partial differential equations via optimization-based implicit feature tracking". en. In: *Journal of Computational Physics* 473 (), p. 111739. ISSN: 0040-1625. DOI: `https://doi.org/10.1016/j.jcp.2022.111739`. URL: `https://www.sciencedirect.com/science/article/pii/S0021999122008026` (visited on 2023).

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